

Technology #149

Implement multi-sensory monitoring system
using X-ray imaging, optical sensors,
acoustic sensors, and spectroscopy within the
battery pack to detect early thermal runaway

Engineering Technology-Transfer Dossier

Initiation
CONFIDENTIAL

Technology Maturity: ENGINEERING_DEFINITION

Generated: 2026-08-30

Not physically validated. Not transfer-ready.

BUYER DECISION PAGE

WHAT IS IT?

Implement multi-sensory monitoring system using X-ray imaging, optical sensors, acoustic sensors, and spectroscopy within the battery pack to detect early thermal runaway initiation

WHY DOES IT MATTER?

Documented recurring problem 'traction battery thermal runaway initiation' in electric vehicles: 5 of 25 NHTSA owner complaints retrieved live at benchmark time (2021–2021) involve fire or battery-related faults. Narratives: 'The high voltage battery on my 2021 Tesla Model 3 Standard Range suddenly failed without any prior warning or gradual degradation. The vehicle was consistently achieving approximately 180 miles of range per full charge. '; 'This is my fourth high voltage battery failure on this vehicle. Battery replacements occurred at approximately 13,000 miles, 60,000 miles, 96,000 miles, and now at 121,000 miles. The first failure occurred at 13,000 mile'. Energy R&D; context: 10 DOE OSTI records on thermal-runaway propagation. LIMITS STAMPED INLINE: voluntary unverified reports, causality UNVERIFIED, counts are NOT incidence (Art. XXI.5 generalized).

WHAT IS ACTUALLY ESTABLISHED?

Engineering definition with 3 governing equations, 11 design inputs, 6 failure modes. Computational model exists. No physical prototype.

WHAT IS NOT ESTABLISHED?

Physical validation, manufacturing process, regulatory pathway, IP ownership.

WHY MIGHT A BUYER CARE?

Opportunity in acoustic. 9-step build plan with specific test articles and acceptance criteria. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHAT DOES THE BUYER GET?

Buyer receives: canonical invention specification (with epistemic classes).; Buyer receives: engineering specification (this document, all blocks classed).; Buyer receives: design input/output graph with explicit IDs.; Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).; Buyer receives: killer-experiment definition and its EIG basis.; Buyer receives: evidence custody references and hashes.; Buyer receives: unresolved-uncertainty register.; Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

WHAT DOES THE BUYER HAVE TO BUILD?

Buyer must create: actual geometry and detailed design.; Buyer must create: sourced engineering parameters and acceptance thresholds.; Buyer must create: physical prototypes.; Buyer must create: all verification and validation results.; Buyer must create: manufacturing process qualification.; Buyer must create: regulatory pathway determination and submissions.

WHAT IS THE NEXT DECISIVE EXPERIMENT?

first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

WHAT WOULD MAKE US KILL IT?

killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT TRANSACTION COULD MAKE SENSE?

Sponsored Validation

What Exists Today

Existing:

- + Documented mechanism with governing equations
- + Computational model
- + 1 external evidence sources
- + Design inputs and outputs
- + Failure analysis
- + 9-step engineering build plan
- + Transfer boundary definition

Does Not Yet Exist:

- Physical prototype
- Qualified manufacturing process
- Clinical data
- Verified IP ownership
- Regulatory clearance
- Commercial product

Buyer Can Commission:

- > Specific bench experiment (see build plan)
- > Prototype fabrication
- > Materials study
- > Manufacturing feasibility study
- > Regulatory pre-submission

Development / De-risking Ladder

CURRENT STATE: ENGINEERING_DEFINITION

NEXT ENGINEERING STEP: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

NEXT VALIDATION STEP: Bench prototype validation per build plan

NEXT REGULATORY STEP: Pre-submission to FDA for pathway classification

COMMERCIALIZATION GATE: Clinical validation evidence required

TRANSFER GATE: Transfer-ready when physical validation + regulatory pathway + IP verified

Engineering Investment Ladder

Next \$5K action: WP-01 - first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed) - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: physical bench/clinical experiment — NOT simulation. Uncertainty: Effort estimate, not quoted.

Next \$25K action: WP-03 - article for VF-001 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: Simulate thermal runaway in a controlled environment with the monitoring system activated to verify detection occurs before propagation t.... Uncertainty: Effort estimate, not quoted.

Next \$100K action: WP-05 - article for VF-003 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: impedance-contrast detection threshold in phantom. Uncertainty: Effort estimate, not quoted.

Top 5 Buyer Diligence Questions

Q1: Is the mechanism physically credible for the intended application?

The governing model includes 3 equations. Key assumptions: sound speed is medium- and temperature-dependent — source the value for the actual medium; alpha is frequency- and medium-dependent; two-way path doubles the exponent; oblique incidence and attenuation modify detectability; AC-002 (Wave speed relation) is applicable only under: non-dispersive propagation in the medium of interest.; AC-003 (Exponential attenuation) is applicable only under: homogeneous lossy medium; single frequency.; AC-001 (Acoustic impedance contrast (pulse-echo)) is applicable only under: normal incidence, plane wave, lossless media.; Model input variables requiring sourced values before any numeric result: c (m/s), f (Hz), A0 (-), alpha (1/m), d (m), Z1 (Rayl), Z2 (Rayl) — all UNKNOWN (no sourced value exists; Art. VIII).; Model output variables produced by the governing model: lambda (m), A (-), R (-) — UNKNOWN until the inputs are sourced and the model is evaluated.; Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): operating frequency [f, Hz] value_status=UNKNOWN; element aperture [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; matching layer Z [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; excitation voltage and duty [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; detection threshold [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; coupling-path attenuation [alpha, 1/m] value_status=UNKNOWN.. UNKNOWN: whether assumptions hold in vivo.

Q2: What evidence supports the core claim?

1 external evidence sources. All are external precedent, not physical validation of this specific invention.

Q3: What is the regulatory pathway?

UNKNOWN. Not yet determined. Requires pre-submission to FDA. No predicate identified for some packages.

Q4: Who owns the IP?

UNKNOWN. IP ownership not verified. Requires legal counsel engagement.

Q5: What would it cost to validate?

First experiment: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed). Estimated effort from build plan. Full validation cost: COST TO BE QUOTED.

Commercial Interest

TARGET APPLICATION: CSF shunt systems for hydrocephalus management

CURRENT ALTERNATIVE: Standard fixed-pressure or programmable CSF shunt valves

MECHANISM DIFFERENCE: Advanced characterization techniques including X-ray imaging, optical sensing, acoustic field imaging, and spectroscopy to study thermal runaway (TR) mechanism

POTENTIAL VALUE DRIVER: Addresses: Documented recurring problem 'traction battery thermal runaway initiation' in electric vehicles: 5 of 25 NHTSA owner complaints retrieved live at benchmark time (2021–2021) involve fire or battery-related faults. Narratives: 'The high voltage battery on my 2021 Tesla Model 3 Standard Range suddenly failed without any prior warning or gradual degradation. The vehicle was consistently achieving approximately 180 miles of range per full charge. '; 'This is my fourth high voltage battery failure on this vehicle. Battery replacements occurred at approximately 13,000 miles, 60,000 miles, 96,000 miles, and now at 121,000 miles. The first failure occurred at 13,000 mile'. Energy R&D; context: 10 DOE OSTI records on thermal-runaway propagation. LIMITS STAMPED INLINE: voluntary unverified reports, causality UNVERIFIED, counts are NOT incidence (Art. XXI.5 generalized).

ADOPTION BARRIER: Requires physical validation, regulatory clearance, and manufacturing qualification

DEVELOPMENT BARRIER: No physical prototype exists; manufacturing process not established

Licensee Capability Fit

IDEAL BUYER: domain-appropriate technology buyer (to be identified by portfolio owner)

REQUIRED ENGINEERING CAPABILITY: Catheter design, fluid mechanics / domain-specific engineering

REQUIRED MANUFACTURING CAPABILITY: Medical tubing extrusion, injection molding, cleanroom assembly

REQUIRED REGULATORY CAPABILITY: FDA 510(k) or PMA submission experience

REQUIRED CAPITAL: Validation: \$50-200K. Full development: \$500K-2M (estimates, not quoted)

REQUIRED MARKET ACCESS: Neurosurgery market channels; hospital purchasing relationships

1. Technology Description

Domain: acoustic

Disciplines: acoustics, transducer engineering, signal processing, materials science

Architecture: PROPOSED architecture (no physical system exists): transducer element + matching layer; excitation / receive electronics; signal chain (detection criterion); coupling path to target medium — realizing: Implement multi-sensory monitoring system using X-ray imaging, optical sensors, acoustic sensors, and spectroscopy within the battery pack to detect early thermal runaway initiation

Subsystems:

?: transducer element + matching layer - ?

?: excitation / receive electronics - ?

?: signal chain (detection criterion) - ?

?: coupling path to target medium - ?

2. Mechanism Architecture

Physical Changes: PROPOSED: Implement multi-sensory monitoring system using X-ray imaging, optical sensors, acoustic sensors, and spectroscopy within the battery pack to detect early thermal runaway initiation

Key Physics: Acoustics / ultrasonics governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

3. Governing Engineering Model

Summary: Acoustics / ultrasonics governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Equations:

AC-002: Wave speed relation. Expression: $c = f * \lambda$. Applicability to this invention: CONDITIONAL; holds under: non-dispersive propagation in the medium of interest. Assumptions: sound speed is medium- and temperature-dependent — source the value for the actual medium. Engaged variables: NOT ESTABLISHED.

AC-003: Exponential attenuation. Expression: $A = A_0 * \exp(-\alpha * d)$. Applicability to this invention: APPLICABLE; holds under: homogeneous lossy medium; single frequency. Assumptions: α is frequency- and medium-dependent; two-way path doubles the exponent. Engaged variables: d .

AC-001: Acoustic impedance contrast (pulse-echo). Expression: $R = ((Z_2 - Z_1) / (Z_2 + Z_1))^2$. Applicability to this invention: CONDITIONAL; holds under: normal incidence, plane wave, lossless media. Assumptions: oblique incidence and attenuation modify detectability. Engaged variables: NOT ESTABLISHED.

Assumptions:

- sound speed is medium- and temperature-dependent — source the value for the actual medium
- α is frequency- and medium-dependent; two-way path doubles the exponent
- oblique incidence and attenuation modify detectability
- AC-002 (Wave speed relation) is applicable only under: non-dispersive propagation in the medium of interest.
- AC-003 (Exponential attenuation) is applicable only under: homogeneous lossy medium; single frequency.
- AC-001 (Acoustic impedance contrast (pulse-echo)) is applicable only under: normal incidence, plane wave, lossless media.
- Model input variables requiring sourced values before any numeric result: c (m/s), f (Hz), A_0 (-), α (1/m), d (m), Z_1 (Rayl), Z_2 (Rayl) — all UNKNOWN (no sourced value exists; Art. VIII).
- Model output variables produced by the governing model: λ (m), A (-), R (-) — UNKNOWN until the inputs are sourced and the model is evaluated.
- Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): operating frequency [f , Hz] value_status=UNKNOWN; element aperture [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; matching layer Z [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; excitation voltage and duty [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; detection threshold [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN;

coupling-path attenuation [alpha, 1/m] value_status=UNKNOWN.

Boundary Conditions:

- to be established by the first bench characterization (not yet performed)

Failure Regimes:

- {'regime': 'battery thermal runaway initiation', 'basis': 'SOURCE_FACT (problem.json)'}
- {'regime': 'detection threshold missed at depth (attenuation)', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'crosstalk between elements corrupting detection', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'resonance drift with temperature', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'bond-wire fatigue under cyclic excitation', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'coupling loss at interface deairing failure', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}

4. Design Inputs

[illegible]

5. Design Outputs

ID	Description	Status	Missing Inputs
DO-001	transducer element + matching layer realizing Implemented CONCEPTUAL	CONCEPTUAL	designed using X-ray diffraction, optical sensors, acoustic sensors, and spectroscopy
DO-002	excitation / receive electronics realizing Implemented CONCEPTUAL	CONCEPTUAL	detailed design of imaging of optical sensors, acoustic sensors, and spectroscopy
DO-003	signal chain (detection criterion) realizing Implemented CONCEPTUAL	CONCEPTUAL	detailed design of imaging of optical sensors, acoustic sensors, and spectroscopy
DO-004	coupling path to target medium realizing Implemented CONCEPTUAL	CONCEPTUAL	detailed design of imaging of optical sensors, acoustic sensors, and spectroscopy
DO-005	transducer element + matching layer interfaces with excitation / receive electronics	CONCEPTUAL	test signal, transducer (all hands off, to implement multi-sensory monitoring system)
DO-006	excitation / receive electronics interfaces with signal chain (detection criterion)	CONCEPTUAL	signal face, impedance, bandwidth, tolerances, multi-sensory monitoring system
DO-007	signal chain (detection criterion) interfaces with coupling path to target medium	CONCEPTUAL	signal face, gain, fidelity, bandwidth, tolerances, multi-sensory monitoring system
DO-008	Device boundaries: patient/fluid inlet and outlet termination	CONCEPTUAL	control/data points required by telemedicine, health devices
DO-009	Flow/force/signal path geometry realizing the mechanism	UNKNOWN	multi-sensory monitoring system using X-ray diffraction, optical sensors, acoustic sensors, and spectroscopy
DO-010	Operating range for operating frequency within which the mechanism holds its stated behavior	UNKNOWN	computed bounds for operating frequency; verification of range
DO-011	Operating range for element aperture within which the mechanism holds its stated behavior	UNKNOWN	computed bounds for element aperture; verification of range
DO-012	Operating range for matching layer Z within which the mechanism holds its stated behavior	UNKNOWN	computed bounds for matching layer Z; verification of range
DO-013	Operating range for excitation voltage and duty within which the mechanism holds its stated behavior	UNKNOWN	computed bounds for excitation voltage and duty; verification of range
DO-014	Operating range for detection threshold within which the mechanism holds its stated behavior	UNKNOWN	computed bounds for detection threshold; verification of range
DO-015	Operating range for coupling-path attenuation within which the mechanism holds its stated behavior	UNKNOWN	computed bounds for coupling-path attenuation; verification of range
DO-016	No closed-loop control is proposed: the invention operates as open-loop as specified; this record exists so the omission is explicit and auditable	CONCEPTUAL	
DO-017	Information path: sensing element -> decision/monitor	CONCEPTUAL	act (minimal); rich data paths are NOT APPLICABLE to this mechanical/physical invention
DO-018	Fixture enabling: impedance-contrast detection threshold	UNKNOWN	rig concept named, build specifications UNKNOWN; acceptance instrumentation
DO-019	Fixture enabling: crosstalk and noise floor characterization	UNKNOWN	rig concept named, build specifications UNKNOWN; acceptance instrumentation
DO-020	Fixture enabling: durability under cyclic excitation — rig concept	UNKNOWN	named, build specifications UNKNOWN; acceptance instrumentation
DO-021	Fixture enabling: phantom detection-threshold mapping	UNKNOWN	rig concept named, build specifications UNKNOWN; acceptance instrumentation

6. Critical Design Parameters

Name	Value	Unit	Basis	Verification
operating frequency	UNKNOWN (no sourced value)	Hz	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
element aperture	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
matching layer Z	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
excitation voltage and duty	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
detection threshold	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
coupling-path attenuation	UNKNOWN (no sourced value)	1/m	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method

7. Failure Modes

Mode	Mechanism	Mitigation	Verification	Residual Uncertainty
battery thermal runaway	Failure by battery thermal runaway	design critical to the ESS failure mode (requires design work)	UNKNOWN (no sourced value)	design assigned candidate value and verification method
detection threshold miss	Failure by depth (attenuation) apertures and duty cycle (power) attempt at getting through the threshold despite coupling gain with target (Antenna)	design assigned candidate value and verification method	UNKNOWN (no sourced value)	design assigned candidate value and verification method
crosstalk between elements	Failure by crosstalk between elements isolating detection	design assigned candidate value and verification method	UNKNOWN (no sourced value)	design assigned candidate value and verification method
resonance drift with temperature	Failure by resonance drift with temperature	design assigned candidate value and verification method	UNKNOWN (no sourced value)	design assigned candidate value and verification method
bond-wire fatigue under cyclic excitation	Failure by bond-wire fatigue under cyclic excitation	design assigned candidate value and verification method	UNKNOWN (no sourced value)	design assigned candidate value and verification method
coupling loss at interface	Failure by coupling loss at interface	design assigned candidate value and verification method	UNKNOWN (no sourced value)	design assigned candidate value and verification method

8. Failure Analysis

Failure Mode	Mechanism	Mitigation	Residual Uncertainty
battery thermal runaway initiation	Failure by battery thermal runaway initiation	design device NOT ESTABLISHED (not needed, triggering not needed, operation mode)	Not needed, triggering not needed, operation mode
detection threshold missed	Failure by detection threshold missed at depth (at end of frequency selection)	depth (at end of frequency selection) on 10000 Hz, detection threshold at 10000 Hz	depth (at end of frequency selection) on 10000 Hz, detection threshold at 10000 Hz
crosstalk between elements	Failure by crosstalk between elements	crosstalk (at end of frequency selection)	crosstalk (at end of frequency selection)
resonance drift with temperature	Failure by resonance drift with temperature	resonance drift with temperature compensation	resonance drift with temperature compensation
bond-wire fatigue under cyclic excitation	Failure by bond-wire fatigue under cyclic excitation	bond-wire fatigue under cyclic excitation	bond-wire fatigue under cyclic excitation
coupling loss at interface	Failure by coupling loss at interface	coupling loss at interface	coupling loss at interface

9. Verification Strategy

ID	Requirement	Method	Acceptance	Result
VF-001	Simulate thermal runaway in a controlled environment	Simulate thermal runaway in a controlled environment	Simulate thermal runaway in a controlled environment	Simulate thermal runaway in a controlled environment
VF-002	run killer experiment: falsification_test	run killer experiment: falsification_test	run killer experiment: falsification_test	run killer experiment: falsification_test
VF-003	impedance-contrast detection threshold	impedance-contrast detection threshold	impedance-contrast detection threshold	impedance-contrast detection threshold
VF-004	phantom detection-threshold mapping	phantom detection-threshold mapping	phantom detection-threshold mapping	phantom detection-threshold mapping
VF-005	resonance tracking across the temperature	resonance tracking across the temperature	resonance tracking across the temperature	resonance tracking across the temperature
VF-006	accelerated cyclic-excitation run-out	accelerated cyclic-excitation run-out	accelerated cyclic-excitation run-out	accelerated cyclic-excitation run-out
VF-007	transmitted-energy check with deliberate heating	transmitted-energy check with deliberate heating	transmitted-energy check with deliberate heating	transmitted-energy check with deliberate heating

10. Validation Strategy

ID	Requirement	Method	Acceptance	Result
VA-001	end-use validation of verified design	NOT ESTABLISHED (defined after verification)	NOT ESTABLISHED (defined after verification)	NOT ESTABLISHED (defined after verification)

11. Materials

Component	Candidate Material	Source	Verification Required	Status
domain candidate material	PZT / PVDF active element	standard ultrasonic transducers	True	ENGINEERING_PROPOSED

12. Bill of Materials

Item	Description	Qty	Type	Material	Supplier	Criticality	Verification
PZT / PVDF active element	domain candidate material	1	NOT ESTABLISHED	PZT / PVDF active element	NOT ESTABLISHED	NOT ESTABLISHED	Reliability verification required

13. Manufacturing

Status: NO MANUFACTURING ANALYSIS POSSIBLE — no design exists; processes are candidates only

Candidate Processes:

- diced-ceramic + backing assembly: domain registry manufacturing route - candidate only; no process qualification exists
- matching-layer lamination and bonding: domain registry manufacturing route - candidate only; no process qualification exists

14. External Evidence

Source 1:

URL :

Excerpt:

15. Transfer Boundary

YOU RECEIVE:

- + Buyer receives: canonical invention specification (with epistemic classes).
- + Buyer receives: engineering specification (this document, all blocks classed).
- + Buyer receives: design input/output graph with explicit IDs.
- + Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).
- + Buyer receives: killer-experiment definition and its EIG basis.
- + Buyer receives: evidence custody references and hashes.
- + Buyer receives: unresolved-uncertainty register.
- + Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

YOU MUST DEVELOP:

- > Buyer must create: actual geometry and detailed design.
- > Buyer must create: sourced engineering parameters and acceptance thresholds.
- > Buyer must create: physical prototypes.
- > Buyer must create: all verification and validation results.
- > Buyer must create: manufacturing process qualification.
- > Buyer must create: regulatory pathway determination and submissions.

YOU MUST VERIFY:

- ? Biocompatibility (ISO 10993)
- ? Sterilization compatibility
- ? Mechanical integrity
- ? Regulatory pathway
- ? IP ownership and freedom to operate
- ? Manufacturing process capability

16. Open Questions

- {'unknown': 'FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-002', 'reason': 'no available verification method measures a quantity family this failure affects (E21-A quantity rule); UNKNOWN disclosure recorded on the row'}
- {'unknown': 'value of critical parameter operating frequency [f Hz]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-001'}
- {'unknown': 'value of critical parameter element aperture [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-002'}
- {'unknown': 'value of critical parameter matching layer Z [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-003'}
- {'unknown': 'value of critical parameter excitation voltage and duty [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-004'}
- {'unknown': 'value of critical parameter detection threshold [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-005'}
- {'unknown': 'value of critical parameter coupling-path attenuation [alpha 1/m]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-006'}
- {'unknown': 'acceptance criterion for verification VF-002', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-002'}
- {'unknown': 'acceptance criterion for verification VF-003', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-003'}

- {'unknown': 'acceptance criterion for verification VF-004', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-004'}
- {'unknown': 'acceptance criterion for verification VF-005', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-005'}
- {'unknown': 'acceptance criterion for verification VF-007', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-007'}

17. Decisive Next Experiment

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-01	first physical article	maintaining the proposed physical verification frequency	impedance contrast	NOT ESTABLISHED	fail to build a test article	100 hrs (to be quoted)
WP-02	constraint-compliance envelope article	per method	constraint-compliance	NOT ESTABLISHED	constraint compliance	100 hrs (to be quoted)
WP-03	article for VF-001	per method	Simulate thermal run	NOT ESTABLISHED	VF-001 verification	100 hrs (to be quoted)
WP-04	article for VF-002	per method	run killer experiment	NOT ESTABLISHED	VF-002 verification	100 hrs (to be quoted)
WP-05	article for VF-003	per method	impedance-contrast	NOT ESTABLISHED	VF-003 verification	100 hrs (to be quoted)
WP-06	article for VF-004	per method	phantom detection threshold	NOT ESTABLISHED	VF-004 verification	100 hrs (to be quoted)
WP-07	article for VF-005	per method	resonance tracking	NOT ESTABLISHED	VF-005 verification	100 hrs (to be quoted)
WP-08	article for VF-006	per method	accelerated cyclic-ex	NOT ESTABLISHED	VF-006 verification	100 hrs (to be quoted)
WP-09	article for VF-007	per method	transmitted-energy ch	NOT ESTABLISHED	VF-007 verification	100 hrs (to be quoted)

18. Buyer Decision Framework

WHY INVEST: Engineering definition exists with 3 governing equations and a 9-step build plan. The next experiment is clearly specified. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHY WAIT: No physical prototype exists. Regulatory pathway is unresolved. IP ownership is unverified. Manufacturing process is not qualified.

WHY REJECT: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT WOULD CHANGE THE DECISION: A successful bench experiment demonstrating the core mechanism. A qualified regulatory pathway. A confirmed IP position.

19. Kill Condition

KILL_IF: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

DISCLOSURE

This dossier distinguishes established evidence, external precedent, computational/modelled results, engineering proposals, and unresolved questions. The absence of physical validation should not be interpreted as evidence that the technology will fail; it means the relevant proposition has not yet been established experimentally.

This dossier does not constitute legal, patentability, FTO, regulatory, or investment advice.